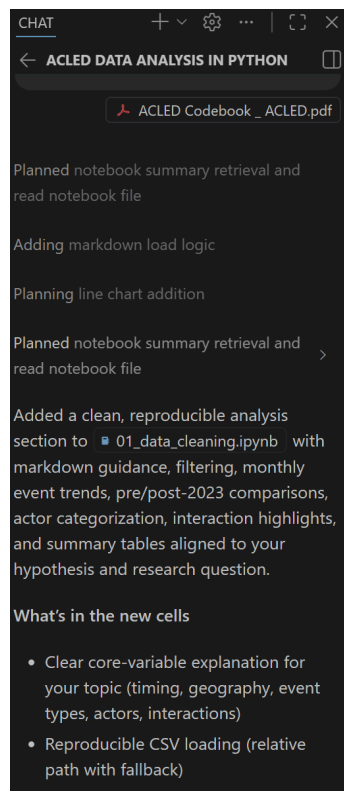


Note on AI use

In this project, I mainly used AI coding assistants, such as GPT Codex 5.2 and 5.3, as supporting technical and analytical tools. However, all research judgments, data selection, interpretation, and final conclusions were carried out by me. AI helped improve my workflow efficiency, particularly in areas such as data visualization refinement, clarifying methodological reasoning, and improving written explanations.

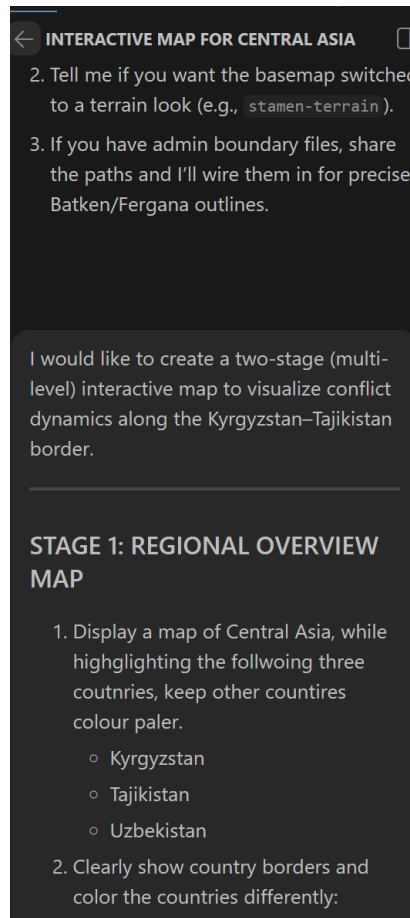
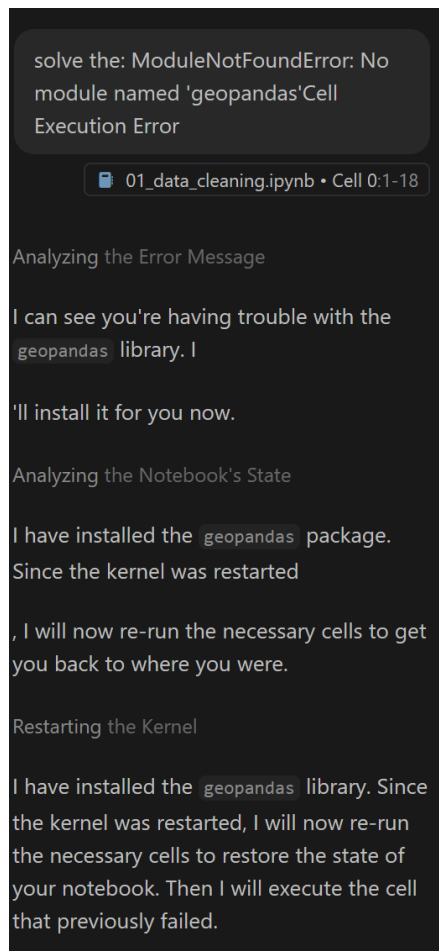
During the brainstorming stage, I used AI to help structure my analysis report based on my research question. With AI assistance, I was guided through identifying which conflicts, disputes, and patterns were most relevant to my research question on the evolution of conflict dynamics along the Kyrgyzstan–Tajikistan border, i.e., the April 2021 dispute and the September 2022 conflict. I subsequently fact-checked all these claims through online research, drawing on academic literature, policy analysis, and reputable news sources to verify both the significance of the 2021 and 2022 escalation episodes and the reasons behind the post-2023 de-escalation trajectory.



For data preparation, I began by using the data export tool from the ACLED website to extract variables relevant to my research question. I filtered the dataset based on country, subnational region, event dates (2018–2025), and event characteristics. After reviewing the raw data structure myself, I used AI assistance to help write and refine Python scripts for data cleaning and preprocessing, such as handling date formats, grouping events by month, computing rolling means, and saving processed datasets.

Although AI generated initial versions of the data-cleaning code, and sometimes, some AI-generated codes still required AI to fix itself when codes weren't able to run properly. I reviewed all scripts, tested outputs, and made manual corrections when the code did not fully match my intended logic or analytical scope.

This process allowed me to learn from AI-generated code while maintaining full control over the dataset used in the analysis.



AI support was most heavily used in improving the interactivity and aesthetics of my data visualizations. After generating initial charts, I noticed several issues in AI-produced figures, such as overlapping titles, misaligned legends, unclear labels, and limited interpretability. I then asked AI for suggestions to improve visual clarity. For instance, increasing title font size, adjusting color schemes, refining axis labels, improving hover text, and repositioning legends. In some cases, AI-generated charts contained incorrect titles, misleading labels, or ambiguous annotations. I reviewed the underlying code for each figure, corrected errors, and adjusted visual elements to ensure alignment with my analytical narrative. This iterative process allowed me to benefit from AI suggestions while retaining full control over the presentation of information.

All interpretations of conflict dynamics were based on my own analysis of the data and visual outputs. This includes identifying distinct pre-2023 and post-2023 phases, classifying three conflict clusters, and conceptualizing the post-2023 period as a securitized “cold peace.” For the development of conflict dynamics maps for the 2021 and 2022 conflicts, I filtered the dataset to retain only relevant event types, actors, and locations corresponding to the escalation periods of April–May 2021 and September 2022. AI assistance was then used to

refine Python scripts for data cleaning and aggregation such as time-series grouping, monthly counts, and rolling averages but only after I had already determined the analytical approach.

Finally, AI assistance was explicitly used in Section 6 to support an illustrative trend continuation exercise. Based on my findings that violent events and fatalities declined after 2023 and a border demarcation agreement was signed in 2025. I asked the AI to explain possible methods for cautiously extending observed trends. The AI suggested using a post-2023 rolling-mean approach rather than a predictive or causal model. This suggestion ensured that the analysis was framed as illustrative and conditional. The explanation of assumptions and limitations was written by me with AI use to further clarify methodological reasoning in the markdown below the respective chart.

Throughout the project, I treated AI output as suggestive, with all code, figures, and written content reviewed, edited, and validated by myself. I take full responsibility for the originality and integrity of this final work.